ChatGPT

Executive Summary

This report investigates how to **engineer a hybrid AI model** that combines the **deep, hierarchical reasoning** capabilities of the HRM (Hierarchical Reasoning Model) with the **large-scale language understanding/generation** capabilities of modern LLMs. We survey the HRM architecture (its code and paper) and compare it with transformer-based LLMs. We review relevant transformer innovations (e.g. rotary embeddings【17†L963-L971】, multi-query attention, sparse/local attention, flash attention 【25†L780-L788】【28†L31-L39】, mixture-of-experts, retrieval augmentation, etc.) focusing on those that could integrate with or inspire an HRM+LLM fusion. We then propose and compare several fusion architectures (e.g., a separate HRM module with a transformer backbone; multi-scale reasoning inside the transformer; iterative "latent reasoning" loops; adaptive computation; and MoE-based designs). For each, we analyze data flow, compute/memory cost, sharing vs separate components, and practical trade-offs.

We detail the **engineering roadmap**: how high-level Python code orchestrates training, data, and model configurations, while performance-critical operations reside in optimized C++/CUDA/Triton kernels. We design a sample repository layout (models, data, tokenizer, training, inference, kernels, etc.) suitable for both research experimentation and production deployment. We trace the **end-to-end data flow** from raw text to logits (tokenization, BPE/embedding, hidden states, reasoning updates, decoder head, output tokens) with explicit tensor shapes.

We lay out a **training strategy**: large-scale language pretraining (cross-entropy next-token) followed by targeted "reasoning curricula" (symbolic/structured tasks, intermediate supervision, possibly reinforcement/ACT training)【6†L195-L203】【10†L19-L27】. We discuss loss functions (next-token vs auxiliary reasoning/Q-learning losses【17†L1111-L1120】) and multi-stage training pipelines (foundation pretraining, instruction tuning, reasoning specialization, alignment).

We provide a **rigorous cost analysis** (parameters, FLOPs, memory) showing how iterative reasoning loops multiply compute and memory, and how techniques like cached KV or sparse attention might mitigate this. We cover low-level optimizations (FlashAttention【25†L780-L788】, fused kernels, quantization modes) and distributed training (data/model/expert parallelism).

Finally, we propose a concrete **hybrid architecture blueprint** (diagram, component specs, shapes, forward/backward flows) and example pseudo-code in PyTorch (illustrating attention, reasoning loops, C++/CUDA bindings, etc.). Throughout we cite primary sources (papers, repos) for every major technique or claim, distinguishing established research vs experimental ideas vs our proposed design.

What HRM Actually Does

The **Hierarchical Reasoning Model (HRM)** is a specialized neural architecture for *goal-oriented reasoning tasks* (puzzles, planning)【10†L7-L16】【6†L195-L203】. HRM's key idea is **hierarchical, multi-timescale processing**: it consists of two interdependent recurrent modules, a **High-level planner** and a **Low-level executor**, which iterate in a nested loop. The high-level module operates slowly with abstract planning,

while the low-level module performs fast detailed computations. This hierarchical recurrence enables large "computational depth" in a *single forward pass*, without needing explicit intermediate supervision. Critically, HRM uses an *Adaptive Computation Time (ACT)* mechanism: at each step it computes "continue/halt" Q-values to decide dynamically whether more iterations are needed【17†L1105-L1113】.

In practice (as seen in the sapientinc/HRM code), the HRM **"inner" model** embeds inputs (e.g. a puzzle grid) and repeatedly updates two hidden states `z_H` and `z_L` via Transformer-like blocks【17†L1009-L1017】【17†L1088-L1097】. The code first performs several nested no-gradient steps (higher stability), then allows a final gradient-enabled update【17†L1090-L1099】【17†L1105-L1113】. The **output** is taken from the high-level state `z_H` and fed through a language-head (linear layer) for token or action logits, plus a small "Q-head" that predicts whether to continue reasoning【15†L1119-L1122】. The HRM paper reports that with just ~27M parameters, this architecture (trained from scratch on a few thousand examples) outperforms much larger GPT-style models on symbolic reasoning tasks like complex Sudoku and the ARC benchmark【10†L27-L36】【10†L39-L42】. In summary, HRM **"does deep reasoning"**: it learns to execute multi-step algorithms in its forward pass, using internal recurrence and learned halting, rather than relying on explicit chain-of-thought prompts or vast data.

What a Modern LLM Actually Does

A **Large Language Model (LLM)** (e.g. GPT, LLaMA, Mistral) is a (usually decoder-only) Transformer that maps token sequences to next-token probabilities【25†L730-L739】【25†L801-L806】. It consists of many stacked Transformer layers, each performing self-attention plus feed-forward processing. Key characteristics of modern LLMs include:

- **Pretraining on massive corpora:** LLMs are typically trained (unsupervised) on billions of words with a next-token cross-entropy objective【17†L1111-L1113】. This gives them broad language understanding, factual knowledge, and generation ability.
- **Single-pass, feedforward inference:** Given context tokens (and cached KV state from prior tokens), each new token is generated by passing through all layers once (possibly using fast FlashAttention kernels【25†L780-L788】 and rotary pos-embeddings【17†L966-L974】). There is no internal iteration or recurrence across output tokens beyond standard sequential decoding with KV cache.
- **Emergent capabilities:** Despite their simplicity, large LLMs (GPT-3 175B, LLaMA 13B/65B, etc.) can follow instructions, hold conversations, do some code generation and summarization, and exhibit reasoning via chain-of-thought prompting【10†L27-L30】. However, their "reasoning" is emergent and shallow: they lack an explicit algorithmic planning module and often hallucinate. They rely on massive scale and data.

In summary, a modern LLM "does" *general-purpose language modeling*: it takes tokenized text (or code) as input, embeds it, applies many layers of self-attention and feedforward, and outputs logits over the vocabulary (for generation)【17†L1111-L1117】. It excels at pattern-matching text and learned associations, but its internal reasoning is often a single-pass approximation with no explicit high-level planning module.

Architectural Differences

The fundamental difference is **structure vs scale**. HRM introduces *multi-level recurrence and planning* explicitly: two recurrent modules (H and L) operate in a nested loop to compute an answer【10†L27-L35】【17†L1088-L1097】. It explicitly learns to stop reasoning via Q-values【15†L1118-L1122】. In contrast, an LLM uses a **feedforward stack of layers** with no internal halting logic. HRM's internal states evolve until a stop signal, whereas LLMs produce output in a fixed number of layers. HRM is *algorithmic*, suited to tasks where stepwise planning is needed【10†L27-L35】. Standard LLMs are *statistical*, suited to language patterns.

Another difference: HRM's training regime is small-data symbolic tasks, whereas LLMs use large raw text corpora. HRM encodes "puzzles" as token sequences (with special embeddings) and optimizes for final answer, while LLMs optimize for next-token prediction. Integrating the two requires reconciling these: can a unified architecture be trained both on text and on structured reasoning tasks?

Relevant Transformer/LLM Breakthroughs

We now survey key developments in Transformer-based models and related techniques, focusing on those with potential to integrate with hierarchical reasoning:

- **Transformer (Vaswani et al., 2017)**: Introduced the self-attention + feedforward block as the core of modern NLP models. Solves sequence modeling via parallel attention【25†L780-L788】. Basis for all LLMs.
- **Decoder-only LLMs (GPT series)**: Trained on next-token prediction, demonstrating emergent few-shot and instruction-following (GPT-3, GPT-4, etc.)【10†L27-L35】. Useful for general language, code, summarization. Usually use fixed number of layers and compute, with KV caching for generation.
- **BERT-style (Encoder) models**: Bi-directional transformers for embedding text; less directly relevant for generation but influential in architecture (masked LM pretraining).
- **RoPE (Rotary Positional Embeddings)**: A relative-position encoding that rotates query/key projections by position-dependent angles. It simplifies long-range generalization and is widely used in LLMs【17†L966-L974】. Compared to absolute embeddings, RoPE imposes no maximum length and couples well with attention. Useful for LLM+HRM to handle sequences of varying lengths.
- **ALiBi (Attention with Linear Biases)**: A simple method that adds a fixed linear bias to attention scores based on distance (no learned embeddings)【25†L780-L788】. It allows training on short sequences and extrapolating to longer inference sequences without positional embeddings. Might help HRM+LLM handle varying context lengths (though HRM uses learned or RoPE in code).
- **Multi-Query (Single KV) Attention**: In generation, using multiple query heads but only one shared key/value head greatly speeds inference by halving KV memory bandwidth【28†L15-L24】【28†L29-L38】. Many LLMs (PaLM, GPT-3) adopt this. However, MQA can degrade quality. **Grouped-Query Attention (GQA)** interpolates: split queries into G groups, each with its own KV head【28†L15-L23】【28†L52-L58】. GQA can approach full multi-head quality with near-MQA speed. For HRM+LLM, these attention variants determine decoder speed vs quality trade-offs.
- **FlashAttention**: A CUDA kernel (FlashAttention) that fuses query-key-value matmuls with softmax and uses tiling to reduce memory I/O【25†L780-L788】. Provides large speedups on GPUs for

Transformer attention, especially on longer sequences. We would use this for both HRM (recurrent loops) and LLM parts to maximize throughput【25†L780-L788】.

- **Sparse/Local Attention** (Transformer-XL, Longformer, BigBird): Techniques to attend only to local windows or sparse patterns to scale to long contexts. Transformer-XL uses segment recurrence and relative pos encoding to break fixed-length context【37†L23-L32】. Hierarchical reasoning might benefit from similar ideas: e.g. local attention for fast updates in LLM blocks, or maintaining hidden state between reasoning iterations.
- **Mixture-of-Experts (MoE)**: Models like Switch Transformers/GLaM route different tokens to different expert sub-networks to massively scale parameter count without proportional compute【28†L31-L39】. In an HRM+LLM, one could allocate experts for different reasoning or knowledge sub-tasks (e.g., one set of layers expert at math reasoning, another at commonsense). However, MoE introduces complexity (load balancing, routing) and may conflict with a tight hierarchical loop.
- **KV Cache / Generation Optimizations**: Caching past keys/values (standard in Transformers) speeds autoregressive generation. Speculative decoding (Li et al. 2025) leverages multiple partial outputs to sample tokens faster【34†L24-L33】. These could apply to HRM if we alternate reasoning with token generation: e.g. do partial decode, then reasoning steps, then continue decode.
- **Quantization & Distillation**: Techniques like INT8/4 quantization, knowledge distillation (e.g. DistilBERT) reduce model size and speed up inference. For an HRM+LLM, quantizing large static parts (the LLM) could save memory, while keeping reasoning modules in higher precision.
- **Retrieval Augmentation (RAG)**: Augmenting a model with an external memory or retriever to fetch relevant documents【Lewis et al.】. This isn't core to the architecture, but is a way to expand knowledge without massive parameters. An HRM+LLM could use RAG to retrieve facts or subgoals during reasoning.
- **Recursive & Adaptive Computation**: Techniques like Graves' ACT (Adaptive Computation Time) allow a model to use a variable number of steps per input【17†L1105-L1113】. HRM inherently uses an ACT-like mechanism (halting Q-values). Similar ideas appear in models like LayerDrop or Universal Transformers (learned depth). Such methods are directly relevant: they let the model "think more" on hard problems.
- **Chain-of-Thought & Latent Reasoning**: Recent work shows that prompting LLMs to output intermediate steps (chain-of-thought) improves reasoning. HRM embeds reasoning internally, but one could imagine explicitly training intermediate states or using a secondary module to process the transformer's hidden states iteratively (e.g. *"Recurrent Transformer"* approaches)【10†L27-L35】.
- **Other Efficiency Tricks**: Rotary pos-emb, ALiBi, attention kernel optimizations (e.g. Triton kernels), low-rank factorizations, etc. These reduce memory or speed up computation and can help if we stack reasoning loops on top of LLM layers.

In each case we must ask: *Is this useful for HRM+LLM?* For example, **adaptive attention** (ACT style) is directly in HRM. **Sparse attention** may help long-context language processing. **MoE** might allow specialized reasoning experts, but complicates integration. **FlashAttention** and **multi-query** purely optimize existing compute without changing capabilities. We will draw on these advances to balance the complexity and cost of our hybrid.

Candidate HRM+LLM Fusion Architectures

We explore several high-level fusion designs. Each blends **structured reasoning** (HRM-style) with **broad language modeling** (LLM):

- **Architecture A – Separate HRM Reasoning Module + Transformer Backbone:**
  A standard pretrained LLM handles all language input/output. A separate **HRM-like module** takes as input the LLM's hidden states (or raw input tokens) and performs explicit reasoning. For example, given a question in text, the LLM first encodes it; then an HRM module (or an ACT-wrapped recurrent subnetwork) iteratively processes that encoding to refine a plan or intermediate answer; finally, the LLM decoder uses this to generate the output.

- *Components:* LLM encoder/decoder (Transformer); HRM planner (two-level recurrent net).

- *Data flow:* [Text→Tokenizer→LLM encoder→(sequence of hidden states)] → *HRM reasoning* → final state → LLM decoder → text output.
- *Tensor flow:* LLM produces context vector `h`; HRM consumes `h` (possibly repeated across its time steps) and outputs a refined `h'`; decoder attends on `h'`. If iterative, HRM may also feed intermediate signals back to LLM layers via cross-attention.
- *Pros:* Keeps language and reasoning modules distinct. LLM can remain general-purpose and pretrained. HRM can be smaller and specialized. Easier to train (train HRM as an add-on).
- *Cons:* Integration difficulty: how to connect HRM's output back to the LLM? Inference may require back-and-forth between modules. Redundancy: both modules have some self-attention layers, etc. Possibly inefficient (two separate networks).
- *Compute:* LLM cost (big) + HRM cost (smaller, but repeated). Memory: KV cache for LLM, plus HRM state.
- *Sharing:* Could share the token embedding between LLM and HRM. Might share some attention blocks if architecture matched. But likely separate.

- *Difference from normal Transformer:* Introduces a second recurrent loop outside the main transformer stack.

- **Architecture B – Hierarchical Reasoning inside the Transformer:**
  Incorporate hierarchical/recurrent structure **within each transformer block** or across blocks. For instance, alternate layers designated as "high-level" and "low-level" reasoning layers, or insert HRM-like loops inside blocks. One could design a Transformer where every few layers, there is an inner loop that refines hidden states.

- *Components:* Standard Transformer layers, but re-organized (e.g. group layers into "stages"). Possibly a top-level slower pathway and a faster residual pathway.

- *Data flow:* Input tokens → initial layers → hierarchical loop (e.g. high-layer activation feeds into a set of recurrent sub-layers, then combines back) → remaining layers → output.
- *Tensor flow:* Hidden states move through layers; at certain points, recurrent blocks take them and iteratively update (like Universal Transformer ideas).

- *Pros:* Deep integration; potential for end-to-end training. No completely separate module, so all parameters can influence each other.
- *Cons:* Very complex to design and train. Hard to reuse off-the-shelf pretrained LLM weights. Recurrent elements may break parallelism (though could unroll a fixed number of steps).
- *Compute:* Potentially much higher (sub-graph repeated multiple times). Must carefully control loops (maybe using ACT to halt).
- *Memory:* Intermediate recurrent states could grow large.
- *Sharing:* Fully integrated, so components share everything.

- *Difference:* The transformer itself becomes dynamic and multi-scale, not just a fixed stack.

- **Architecture C – Latent Reasoning Loop (Iterative Hidden-State Refinement):**
  Allow the model to internally "think" by performing *additional hidden-state updates* between token generations. For example, after encoding the prompt, the model may run extra Transformer or HRM iterations on its final hidden state (without emitting tokens) before decoding. Or after generating each token, run a few self-conditioning updates on the hidden state. This can be seen as a form of deep thinking between words.

- *Components:* A controller that decides to run extra updates on the latent state (could be a loop of standard Transformer blocks or a special reasoning block).

- *Data flow:* Text → embed → LLM feedforward → (optionally run additional "reasoning" steps on final hidden state) → output logits.
- *Tensor flow:* One can imagine inserting a function `z_new = f(z_old)` (where `f` might be an HRM block or a few transformer layers) and then using `z_new` for the next output.
- *Pros:* Simple extension: reuse LLM layers or a small reasoner. No change to token flow. The model itself decides (via a learned policy) when to stop.
- *Cons:* Training such latent loops is tricky (no direct supervision of intermediate states). Might require specialized loss (e.g. ACT-style or consistency objectives). Also slows inference.
- *Compute:* Each reasoning step costs roughly one more transformer pass (or HRM pass). If it does N extra steps, cost ×N.
- *Memory:* Needs to store intermediate hidden states for backprop. Possibly heavy.
- *Sharing:* Could reuse the same weights for each step (like an RNN) or have unrolled copies.

- *Difference:* Adds iterative refinement on the fixed context before/after generation.

- **Architecture D – Adaptive Reasoning / Compute Budget:**
  Build a model that **dynamically chooses its compute** per example. This uses ACT or learned halting. Essentially, the model can "ponder" a variable number of steps. For instance, a Transformer might contain a learnable halting gate after some layers: if the model is confident, it stops and outputs; if not, it continues deeper or runs extra loops. HRM's ACT mechanism is a prime example of this (it computes Q-values to halt)【15†L1118-L1122】.

- *Components:* At one or more points in the architecture, an algorithm decides whether to continue computing. Possibly implemented as an extra network (like HRM's Q-head) predicting "done".

- *Data flow:* After each step (e.g. layer or token), evaluate continue/halt. If continue, do more computation; else, output.
- *Tensor flow:* Intermediate states carry along, halted examples are masked out (like HRM does with carry resets).
- *Pros:* Efficiency: easy examples use less compute. The model learns to allocate depth. Could reduce wasteful reasoning on simple prompts.
- *Cons:* Very complex to implement and train (requires RL or differentiable halting). Batching is hard (different tokens may need different depth). HRM's code shows this complexity【15†L1119-L1128】.
- *Compute:* On average less, but worst-case same or more. If some inputs need 50 reasoning loops, it's huge.
- *Memory:* Must track dynamic per-example state (halting, steps). Additional memory for RL targets (if used).

- *Difference:* The model is non-uniform: not a fixed graph but a conditional compute graph.

- **Architecture E – Mixture-of-Experts Reasoning:**
  Use **specialized "experts"** for different sub-tasks or knowledge domains. For example, have an expert network specialized for logical reasoning, another for arithmetic, another for general language, etc. A routing mechanism (possibly trainable) directs parts of the computation to the appropriate expert. The base LLM layers could themselves be an MoE (as in GShard/Switch) or could call separate expert modules.

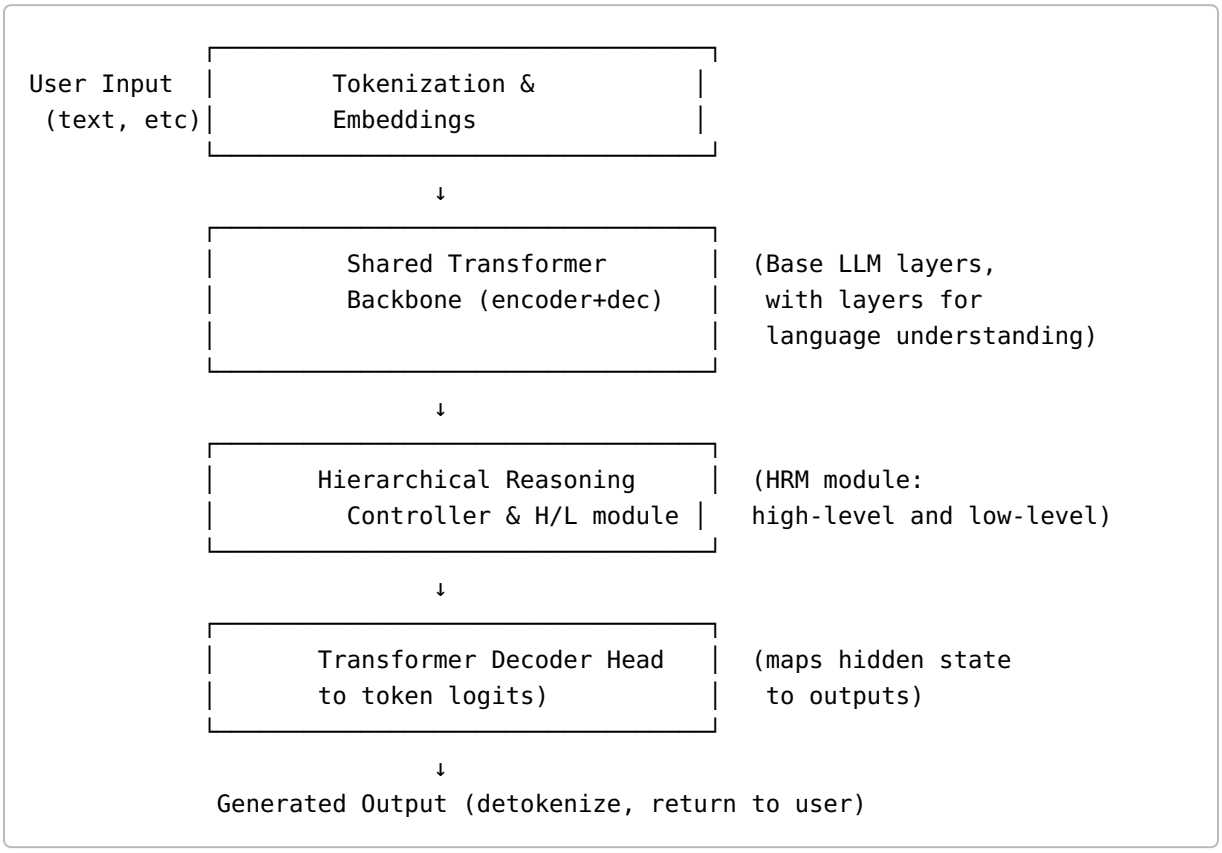
- *Components:* A set of sub-models (experts), each possibly a full HRM or LLM module. A gating network chooses which expert(s) to activate for a given input or token.

- *Data flow:* Input tokens pass through a router (learned MoE gate) that sends them to one or more experts. Those experts process and produce output which is merged. E.g. a question about math might go to a "math expert" HRM module.
- *Tensor flow:* At an MoE layer, the input tensor splits (via gating probabilities) into multiple heads, each sent to an expert transformer; outputs are recombined.
- *Pros:* Potential to scale parameters and diversity without linear compute cost. Can encapsulate HRM-like reasoning in some experts and LLM knowledge in others. Allows specialization.
- *Cons:* Increases system complexity (routing, load balancing). Hard to train reliably (experts can become unbalanced). Might be overkill unless truly multimodal (language+vision+math).
- *Compute:* If top-k gating (k=1 or 2), then each token sees only a few experts, so cost is moderate. But synchronizing experts adds overhead.
- *Memory:* Must store all expert parameters (could be very large). Activations only in a few experts per token.
- *Difference:* Introduces conditional network structure; model is an ensemble of sub-models rather than one flat net.

For each architecture, the **trade-offs** include: integration complexity vs modular clarity, compute vs performance, and ease of leveraging existing weights. Architecture A is most modular but may duplicate effort. B/C are more unified but complex. D/E introduce adaptability at the cost of implementation challenge. In practice, a useful design might combine these: for example, *Architecture C with D*: a latent reasoning loop that uses an ACT halting (so it can do 0,1, or many steps). Or *A with E*: a base LLM plus an MoE of HRM experts. We will compare them concretely later.

Detailed System Architecture

Based on the above, we envision a **hybrid model** with these core components:



This is a variation of **Architecture A** where the base LLM encoder produces hidden states that feed into an HRM module, whose output then guides the decoder. In more detail:

- **Tokenizer/Embedding:** Standard subword tokenizer (e.g. SentencePiece/BPE). Convert text to token IDs, add special tokens, pad, etc. Embedded via an `Embedding(num_tokens, d_model)` layer. Optionally add positional encodings (RoPE or learned). These embeddings feed into the Transformer.
- **Transformer Backbone:** A stack of N layers (each layer: Self-Attn + Feedforward, with residuals) that encodes the input sequence. This could be pretrained (like GPT or LLaMA). The output is a sequence of hidden vectors $H=[h_1,\dots,h_L]$.
- **Hierarchical Reasoning Module (HRM):** Takes H (or some summary of it) as input. Internally has:
- A **low-level RNN** (or fast L-level transformer module) that updates a "detail" state z_L rapidly.
- A **high-level RNN** (slow, with fewer iterations) updating an abstract state z_H.
- These run in an interleaved loop (like the HRM code), possibly with learned halting.
- The HRM's input embeddings come from the transformer's output or initial embedding, possibly concatenated (like the puzzle embedding in HRM code【17†L1009-L1017】). It uses the same attention and feedforward blocks internally (maybe lighter ones). The final z_H is the "reasoning memory".

- **Language Model (Decoder) Head:** A linear layer (or softmax) that maps z_H (or combined with z_L) to output token logits. Alternatively, if using an encoder-decoder, the decoder cross-attends on z_H.
- **Output:** The logits are converted to probabilities (softmax) and tokens. In generation mode, tokens may be fed back for continuation.

Data Flow (Tensor shapes): Suppose input batch is (B, T_in) tokens. After embedding, we have (B, T_in, D). The Transformer backbone produces (B, T_in, D). We may pool or project this to a fixed-size initial state for HRM (e.g. take the first token or use a linear readout), resulting in (B, D). The HRM loop then produces updated states `z_H, z_L` of shape (B, D). After R reasoning steps, we output (for each batch) a vector or sequence of vectors. The decoder head then maps (B, D) to (B, V) logits (V = vocab). In generation, (B, D) extends to (B, T_out, D) sequentially.

Training Flow: We would compute cross-entropy between the final logits and target tokens, plus any auxiliary losses (e.g. if using ACT, a loss on the halting decisions【15†L1118-L1122】). Backprop flows through decoder, through HRM, into the transformer. Because HRM does multiple steps, gradients either need to be back-propagated through the unrolled steps (costly) or we can use the truncated scheme like HRM does (only last step carries grad)【17†L1097-L1105】. We also maintain optimizer state for all parameters (likely using AdamW). Learning-rate schedules, weight decay etc. follow LLM best practices (linear warmup, cosine decay, etc).

Hardware/Parallelism: Training would run on GPUs. The Transformer backbone and HRM blocks use standard matrix multiplies (accelerated on GPUs). We would use FlashAttention kernels for speed【25†L780-L788】. If model is large, we might do data parallelism (DDP) or tensor parallelism (Megatron style). HRM's recurrence means some sequential steps, but these are within a forward pass, so multi-GPU can still parallelize the batch.

Parameter count: If using a medium LLM (say 13B) plus a small HRM (say 50M), total ~13.05B. The HRM component is tiny relative to the LLM. If desired, one could also use smaller base. The HRM paper used only 27M and solved hard tasks【10†L27-L35】, but for language we need larger base.

In essence, this system extends a standard transformer with an **inner reasoning loop**. It is fundamentally different from a normal Transformer because it *recurses* internally: the hidden state z is updated multiple times before producing output【17†L1088-L1097】. This extra depth is the core new capability.

Python vs C/C++/CUDA/Triton Responsibilities

In a modern AI codebase, **Python is the high-level orchestration layer**, while performance-critical ops run in compiled code. Specifically:

- **Python (with PyTorch):**
- Defines the **model architecture**: writing classes for modules (Transformer layers, HRM blocks, embedding, etc.) in Python/PyTorch. For example, the HRM code's `HierarchicalReasoningModel_ACTV1` class is Python/PyTorch【15†L1124-L1133】.
- Loads and preprocesses **data**: reading datasets, tokenization (using a library or custom Python code), batching, augmentation. E.g. HRM's `dataset/build_sudoku_dataset.py`.

- Controls **training loop**: batching, forward pass, loss computation, backward pass (via `torch.autograd`), optimizer step (e.g. AdamW implemented in PyTorch).
- **Experiment configs**: using config files or argument parsing (YAML/Argparse) to set hyperparams (learning rate, architecture variants, etc.).
- **Checkpointing**: using `torch.save` / `torch.load` to save model weights.
- **Inference API**: for deployment, a Python wrapper could load the model and serve queries (e.g. via Flask/FastAPI).
- **Utilities**: logging, evaluation metrics (accuracy, perplexity), W&B tracking as in HRM repo【6†L259-L268】.
- **Distributed training orchestration**: launching `torchrun`, setting up DDP across GPUs/nodes.
- **Experiment management**: launching runs, seeding RNGs, etc.

- **Testing**: writing unit tests in Python for components, integration tests for data pipeline.

- **C/C++**:

- **PyTorch core**: Much of PyTorch's tensor ops (matrix multiply, softmax, etc.) are in optimized C++ (ATen) and CUDA. Python code calls `x @ y`, but the heavy lifting is C++ (cuBLAS, cuDNN) behind the scenes.
- **Custom C++ extensions**: For very custom ops, one writes a PyTorch C++ extension (using `torch::autograd::Function` or pybind11). E.g., a custom normalization or activation not in PyTorch. HRM's code uses `RMSNorm` implemented likely in C/C++ for speed.

- **Binding libraries**: Torchscript JIT is in C++. Many third-party HPC kernels (e.g. optimized BLAS) are C/C++.

- **CUDA/Triton**:

- **Attention kernels**: Custom CUDA kernels (or Triton code) for attention, as in FlashAttention. These do the Q*K and softmax in tile-parallel fashion【25†L780-L788】. FlashAttention is implemented in CUDA (and now also Triton) for speed.
- **Fused operations**: Operations like GELU, layernorm fused with residual, etc., can be written in Triton for efficiency.
- **Low-level tensor ops**: Convolutions, GPU fast math routines.
- **Quantization kernels**: If using int8 inference, specialized CUDA or Triton kernels handle quantized matmuls and dequantize.
- **Memory management**: Allocation of GPU memory, free, etc. Largely handled by PyTorch, but could be tuned in C++.
- **Parallel communication**: NCCL (in C++) for multi-GPU all-reduce in DDP.
- **GPU-specific code**: For example, launching an efficient fused kernel for the HRM's hidden-state loop might use Triton.

Interaction Mechanisms: Python calls these lower-level routines via interfaces: PyTorch's C++/CUDA op registration (dispatch). For instance, a Python `torch.nn.Linear` triggers a C++ CPU matmul or CUDA matmul. For custom ops, one can write a `.cpp` or `.cu` file, compile to a shared lib, and `load` it in Python. Pybind11 is used under the hood by PyTorch to bind custom C++ classes to Python. Triton kernels can be launched by importing a Python module (the Triton library provides a decorator to write kernels in a

Pythonic syntax). In all cases, from Python we just see an op call, but behind it it goes to optimized binary code. For example:

```
class CustomLayer(torch.nn.Module):
    def forward(self, x):
        return torch.ops.my_ops.my_custom_attention(x)  # calls C++/CUDA op
```

The HRM code likely uses standard PyTorch ops (matmul, softmax) which themselves are C++/CUDA. For example, when HRM does `self.self_attn(...)`, that module likely calls into a fused FlashAttention op (in CUDA) via a Python wrapper (FlashAttention has a Python pip package).

In summary, **Python = orchestration, C++/CUDA = compute**. The engineer writes the high-level model logic in Python/PyTorch (defining how tensors flow and what modules exist), but delegates the heavy numeric work (matrix ops, kernels) to lower-level implementations. This way we get both flexibility (Python) and speed (compiled code).

Repository / Folder Structure

A clean codebase separates concerns. An example structure for our HRM+LLM project might be:

```
project_root/
├── configs/               # Experiment YAML/JSON configs (model sizes,
hyperparams)
├── data/                  # Raw/preprocessed data (downloaded corpora, etc.)
├── datasets/              # Scripts to build/stream datasets (tokenization,
filtering)
│   ├── build_text_corpus.py
│   ├── build_reasoning_dataset.py
│   └── ...
├── tokenizer/             # Tokenizer code and vocab
│   ├── train_sentencepiece.py
│   └── tokenizer.model
├── model/                 # Model code
│   ├── transformer/       # Standard Transformer blocks
│   ├── attention/         # Custom attention modules (flash, rotary, etc.)
│   ├── embeddings/        # Embedding layers, positional encodings
│   ├── hrm/               # HRM-specific modules (H/L layers, controller)
│   ├── heads/            # Output heads (language head, Q-head, etc.)
│   └── __init__.py
├── training/              # Training loop, trainer classes
│   ├── train.py
│   ├── validation.py
│   └── distributed.py    # DDP setup
├── inference/             # Inference/evaluation scripts
```

```
│   ├── generate.py
│   └── serve.py          # API for serving model
├── optimization/          # Scripts for quantization, pruning
├── kernels/               # Custom C++/CUDA/Triton code
│   ├── cuda/             # .cu files
│   ├── cpp/              # .cpp/pybind code
│   └── triton/           # Triton kernels
├── checkpoints/           # Saved models and optimizer states
├── scripts/               # Utility scripts (data download, format conversion)
├── tests/                 # Unit and integration tests
└── docs/                  # Documentation, design diagrams, instructions
```

Explanation:

- `configs/`: Centralizes all hyperparameters and options (model size, learning rate, layers, HRM cycles, etc.). Avoids hard-coding values.
- `datasets/` and `data/`: Raw data (e.g. text corpora, puzzle images) in `data/`; scripts to preprocess (clean, dedupe, tokenize) in `datasets/`. Keep data raw and not clutter the repo with large files (use `.gitignore` or pointers).
- `tokenizer/`: Contains code to train/load the tokenizer (e.g. SentencePiece) and store vocab files.
- `model/`: High-level structure of the model. We separate by function: e.g. `transformer/` has generic Transformer encoders/decoders; `hrm/` has the hierarchical reasoning blocks (like the HRM code's modules); `attention/` has variants (FlashAttention wrapper, ALiBi module, etc.); `heads/` has the output layers (language modeling head, optional classification heads).
- `training/` and `inference/`: Logic to run training or inference. These use the model from `model/` and data pipeline. Training code handles loops, logging, saving checkpoints. Inference code loads a checkpoint and runs the model (e.g. greedy or beam search).
- `kernels/`: All custom low-level code. For example, if we implement a new fast attention in Triton, put the kernel `.cu` or `.py` here. Also C++ extension code to wrap custom ops. This isolates GPU-specific code from Python logic.
- `checkpoints/`: (Often not in repo, but we note where to save). We store model weights and optimizer states here, versioned by experiment.
- `scripts/`: Misc utilities, like downloading data or computing metrics.
- `tests/`: Test everything. Ensure no circular imports; use absolute imports for modules. Keep research prototyping code out of production libraries.

Dependency Direction:

- `model/` should not depend on `training/` or `inference/`. Training uses model, not vice versa.
- `dataset/` uses `tokenizer/` and `model/` (for vocab sizes, etc.), but model code should not call dataset-specific scripts.
- `kernels/` are used by `model/` or lower-level ops, but `kernel/` code should not import Python modules outside of PyTorch itself.
- Maintain separation: e.g., HRM architecture code goes in `model/hrm/`, not scattered in training script. This ensures components are modular and testable.

Preventing circular deps:

- Use import conventions (e.g. absolute imports). Avoid putting model logic inside training scripts.
- Any cross-dep: e.g., if both `transformer/` and `hrm/` need an `Attention` class, put it in a shared `attention/` module.

Research vs Production Code:

- Keep experimental code (new ideas, notebooks) out of core modules. These can live in `scripts/` or `notebooks/`.
- Core modules in `model/` should be reusable. For example, we may want to reuse the HRM loop for multiple tasks, so make it generic.

Configuration:

- Use a config system (Hydra, YAML) so that adding a new architecture (variant of HRM or Transformer) is easy via config flags, not rewriting code.

Checkpoints:

- Organize by experiment name / date. Possibly symlink `latest.pt`.
- Ensure checkpoints save only necessary states (model weights, optimizer, RNG states optionally).

Extensibility:

- To add new architectures (e.g. a new attention variant), one should add a new class in `model/attention/` and register it via the config, without touching existing code.

In a **minimal prototype**, many of these directories would collapse. E.g. `model/` might just contain one file, `data/` might not exist (hardcoded toy data), etc. But as we scale, the structure above helps maintain order.

Input and Data Pipeline

We describe the path from raw data to model tensors:

```
Raw text/symbols → cleaning → tokenization → token IDs → embeddings →
(Transformer/HRM) → hidden states → head → logits → token probabilities → output
tokens
```

- **Raw Dataset:** Sources include web text, code, or structured puzzle data. For language tasks: books, websites, Wikipedia, code repos, etc. For reasoning tasks: e.g. ARC puzzles, Sudoku examples. These datasets must be filtered/cleaned (remove bad content, deduplicate)【8†L366-L375】. Use scripts in `datasets/` to parse raw files and output e.g. plain text or JSON entries.

- **Cleaning/Filtering:** Remove undesirable tokens, limit document length, fix encoding errors. Possibly use language detection, remove non-target languages, etc. Also handle **deduplication** by hashing or similarity, as large corpora often have near-duplicates. For multi-lingual, we might segment by language and filter out undesired parts.

- **Tokenization:** Train or use a tokenizer (e.g. SentencePiece) on the preprocessed text. This yields a vocabulary of subword units. For code generation, include code-specific tokens. Add *special tokens* like `<eos>`, `<pad>`, `<s>`, `</s>`, etc. The tokenizer script (`tokenizer/train*.py`) outputs a model file. During training, text is converted to token IDs (length N). Long documents are chunked to the model's max context (e.g. 2048). Use padding/masking for batching variable lengths.

- **Context Length & Position Encoding:** Decide on a maximum sequence length (based on hardware). For positions within that length, use relative (RoPE) or learned embeddings. This defines how positional information is encoded. E.g. HRM code had `pos_encodings: learned` or `"rope"` (config)【17†L963-L971】. ALiBi would add biases instead of embeddings. The pipeline builds an "attention mask" (1s for real tokens, 0s for pad) and possibly `position_ids` if using RoPE.

- **Batch Construction:** Combine sequences into batches. Possibly bucket by length to minimize padding. Use PyTorch `DataLoader` with a custom collate function. For HRM tasks like puzzles, inputs might be images or grids; these must be flattened or encoded as special tokens (HRM repo uses a puzzle dataset script【6†L268-L276】).

- **Data Streaming / Sharding:** For large corpora, use streaming (WebDataset or TFRecords). In multi-GPU setups, shard data so each worker gets distinct subset. Use `torch.utils.data.DistributedSampler` for DDP.

- **Instruction & Reasoning Data:** After base corpus, additional fine-tuning data: instruction prompts, code, reasoning benchmarks. These should be formatted similarly (as text, possibly with instruction tags). Curriculum learning can be applied (e.g. start with easier tasks, progress to harder reasoning).

- **Tokenizer Interface:** After IDs, the first model input is an integer tensor `tokens` of shape (B, T). Optionally a `puzzle_id` or other metadata is also fed (as in HRM's `puzzle_identifiers`) 【17†L1010-L1019】. The model's first layers transform these IDs to embeddings (embedding matrix lookup, giving (B, T, D)).

- **Transformations:** The embeddings are scaled (`*sqrt(hidden_size)`) and positional encodings added. They become the initial hidden states input to the Transformer/H/L modules【17†L1010-L1019】. If using curriculum or multiple sequences, there may be attention masks to prevent cross-contamination.

- **What model receives:** Numerically, it receives a tensor `input_ids` (B×T), maybe `attention_mask` (B×T), and other flags. It outputs a tensor `logits` (B×T×V) (for full sequence modeling) or (B×V) if predicting one token at a time. Internally, it produces hidden states of shape (B×T×D) at each layer. The HRM reasoning might produce an (B×D) vector or updated (B×T×D) sequence.

Example pipeline:

```
Text question: "Solve the Sudoku."
↓
"Cleaned & formatted"
↓
Tokenize → [<s>, "Solve", "the", "Sudoku", ".", </s>]
↓
Token IDs → [01, 500, 20, 1043, 3, 02] (shape [B=1,T=6])
↓
Embedding → (1×6×D) tensor
↓
Transformer encoder + HRM → hidden (1×6×D)
↓
Logits → (1×6×V)
↓
Softmax → probabilities
↓
Next token predicted (e.g. "Start solving").
```

Each stage's tensor shape is as above, ensuring dimensions match (batch first by convention). If modeling as autoregressive (GPT style), the output for time step t is masked from future. Padding tokens have mask=0 so they don't affect attention.

Model Tensor Flow

We detail the tensor transformations stepwise:

1. **Embedding:** Input `input_ids` has shape (B, T). The embedding layer maps each ID to a D-dimensional vector, yielding `E` of shape (B, T, D). If HRM uses special puzzle embeddings, those might concatenate: e.g. embedding = [puzzle_emb, token_emb] → shape (B, T+P, D), then projected back to (B,T,D)【17†L1021-L1030】.

2. **Positional Encoding:** If learned embeddings: add (B, T, D) learned vectors to `E`. If RoPE: compute cos/sin of shape (T, head_dim) and pass separately into attention. HRM code stored a rotary cache up to max seq length【17†L966-L974】.

3. **Transformer Layers:** Each layer takes `(B, T, D)` and returns `(B, T, D)`. Attention: computes Q,K,V each (B, T, D) → (B, num_heads, T, head_dim) then attention scores and output (B, T, D). Feedforward (SwiGLU or GeLU) transforms (B, T, D)→(B, T, D). Residual adds + norm produce (B, T, D). After L layers, output `H` is (B, T, D).

4. **HRM Reasoning Loop:** Suppose we pool `H` to an initial state: e.g. take `h_1` or mean of `H`: a vector (B, D). Call this `z_H0`, and similarly set `z_L0 = z_H0`. Then for `r=1..R`:

5. Inject input: let `inj = z_H(prev) + E` broadcast (B,T,D).

6. Update low-level: `z_L_new = L_level(z_L_prev, inj)` (applying several L-blocks) → (B, T, D) or (B, D) depending on design.

7. Update high-level: `z_H_new = H_level(z_H_prev, z_L_new)` → (B, D).
   Typically, HRM alternates many `z_L = L(z_L, inj)` and occasional `z_H = H(z_H, z_L)` as loops【17†L1090-L1099】. Finally we get `z_H_final` (B,D).
   If using halting: also compute `q = Q_head(z_H_prev)` and decide whether to continue. If halted, break the loop.

8. **Language Model Head:** The final `z_H_final` (and possibly `z_L_final`) is fed through a linear head: e.g. `logits = Linear(z_H_final)` giving shape (B, V). If generating sequences, each position in output sequence is produced one at a time, each output token embedding is fed back (standard autoregressive).

9. **Softmax + Loss:** `logits` through softmax yields (B,V) probability distribution over tokens. For training, compute cross-entropy with target IDs (B) or (B,T). Also include auxiliary losses: e.g. if HRM ACT, compare predicted `q_continue` vs target (maybe a self-supervised target of when to stop) 【15†L1216-L1228】.

10. **Backpropagation:** Gradients flow from loss to the linear head, then through HRM updates (only last iteration has grad), through input to `z_H0`, then through transformer layers back to embeddings to inputs. Because earlier reasoning steps had `torch.no_grad()`, grads only touch the final step by design【17†L1097-L1105】.

In summary, the main tensor shapes are:

- Input IDs: (B,T)
- Embeddings: (B,T,D)
- Hidden states (per layer): (B,T,D)
- HRM states: (B,D) or (B,T,D) depending how we align HRM's output to Transformer tokens.
- Logits: (B,V) or (B,T,V).

This flow shows how raw text is ultimately turned into model output.

Training Pipeline

We propose a **multi-stage training schedule**:

1. **Stage 1: Language Pretraining.** Train the Transformer backbone (without the HRM) on a large text corpus with next-token cross-entropy. This builds general language knowledge. If starting from scratch, do this for many tokens (billions); or load a pretrained LLM checkpoint (GPT, LLaMA weights) to save time. Use standard recipes: AdamW, linear LR warmup, cosine or step decay, batch sizes of thousands, mixed precision (FP16/BF16). Dataset: diverse text (web, books, code). Optionally include synthetic data (backtranslation, etc) for augmentation. Sequence lengths maybe 2048.

2. **Stage 2: Instruction / Task Tuning.** Fine-tune the pretrained LLM on instruction-following data (e.g. human-written instructions + outputs) to align it. Use supervised fine-tuning (cross-entropy). This is standard for ChatGPT-style. It helps the LLM interface.

3. **Stage 3: Reasoning Data Curriculum.** Introduce datasets that require multi-step reasoning. For example, puzzles like ARC, math problems, logic games. This stage trains the **HRM component**. A possible approach: freeze the LLM backbone (or fine-tune lightly) and attach the HRM module; train on paired (input, solution) with cross-entropy on the solution tokens. Provide supervision at the final answer. Optionally use **self-consistency** or backtracking to refine training (like showing multiple reasoning paths). We may also use auxiliary losses: e.g. if HRM does ACT, use a Q-learning loss for the halting signal (the HRM code uses a kind of binary RL without replay【15†L1210-L1219】). Could include intermediate "reasoning state losses" if we can label subgoals.

4. **Stage 4: Integrated Fine-Tuning.** Jointly fine-tune the full model (LLM + HRM) on both language and reasoning tasks. This encourages the modules to cooperate. We might combine losses (some data is pure language, some reasoning). Use a weighted schedule. Possibly use curriculum learning (e.g. gradually more reasoning steps).

5. **Stage 5: Alignment and Preference Optimization.** If this model is interactive, apply RL from human feedback or classification loss to shape the outputs (like RLHF). This tunes final behavior (politeness, no hallucination). Might only apply to the language head.

6. **Stage 6: Compression/Quantization.** Finally, compress for deployment (distill to smaller model, quantize weights to INT8/4). Use knowledge distillation (train a smaller student to mimic the LLM+HRM teacher) and quantize critical layers.

Training Techniques & Details:

- **Batching and Sequence Length:** Use as long context as GPU allows to maximize reasoning context (HRM may need to see full puzzle). However, HRM as coded used mostly short sequences (puzzles flattened). The text LLM uses say 1024–4096 tokens.
- **Loss Functions:** The base objective is always cross-entropy for language tokens. For the reasoning stages, we might add an **ACT loss**: train the halting head via policy gradient or target-value; HRM code uses a target Q-value (the max of halt/continue Q)【15†L1216-L1224】.
- **Optimization:** AdamW with weight decay ~0.1. Clip gradient norm (~1.0). Use fp16 or bf16 (mixed precision). Use gradient accumulation to simulate large batch.
- **Distributed Training:** Use PyTorch DDP (data parallel). For very large LLM, possibly use tensor parallelism (Megatron). HRM part is small, so mostly data parallel.
- **Evaluation:** Monitor language metrics (perplexity) and reasoning accuracy (exact match on puzzle answers). Early-stop if overfitting on small reasoning sets. HRM paper noted instability if overfitting【8†L381-L389】, so use proper regularization.

Reasoning-specific training: We may also consider **reinforcement learning**: if we cannot label all intermediate reasoning steps, treat the final accuracy as a reward. The HRM code already uses an RL-like internal Q head. Alternatively, generate additional pseudo-training data: use the current model to propose chain-of-thought steps and filter out good ones. Or use techniques like Socratic models to generate hints.

In sum, training is staged: first get general capabilities, then gradually shape it into a reasoner by focused tasks. This mimics human curriculum (learn language, then logic, then integrate).

Loss Functions and Learning Objectives

The **primary loss** is *language modeling cross-entropy* (negative log-likelihood of true next token). This trains the Transformer backbone and language head. For reasoning tasks, we still supervise the final output tokens (the solution), so cross-entropy covers both text and answers.

However, for deeper reasoning we may include **auxiliary objectives**:

- **Halting/Q-Loss (ACT):** The HRM ACT mechanism predicts Q-values for "halt" vs "continue" at each reasoning step【15†L1118-L1122】. We can train these with a temporal-difference or policy-loss. The HRM code computes a "target_q_continue" by comparing the next step's Qs【15†L1224-L1230】. The loss could be MSE or binary cross-entropy pushing the model to predict the better action. This encourages the model to learn to stop when it has solved the problem.

- **Intermediate State Consistency:** If possible, penalize the model if it changes hidden states after halting (so last state is fixed). Also, ensure early stop if correct answer reached (supervision from final answer).

- **Distillation:** Train smaller model to match a larger (for scaling down). Or train the language backbone to imitate a teacher's logits on a mixture of LLM+HRM outputs.

- **Contrastive Objectives:** Could use contrastive loss to align embeddings (less relevant here, more for retrieval).

- **Reconstruction / Autoencoding:** If we wanted to ensure reasoning details aren't lost, we might reconstruct sub-questions. But in our fused model, probably not needed.

- **Adaptive Computation Loss:** There are papers (like Graves ACT) that add a penalty for each extra step (e.g. small lambda per step) to discourage needless loops. Could implement a cost for each reasoning iteration to incentivize efficiency.

Overall, the model is trained end-to-end mostly with cross-entropy. HRM architecture is intended to **"discover reasoning"** implicitly through backprop from the final answer loss【10†L27-L35】, without explicit CoT labels. Auxiliary losses are mainly for the adaptive mechanism.

Compute and Memory Analysis

We estimate the **cost** of this hybrid model. Let:

- N_L = number of layers in LLM (e.g. 24).
- D = hidden dimension (e.g. 4096).

- T = sequence length (e.g. 1024).
- B = batch size.
- R = average reasoning iterations (from ACT or fixed schedule).

Parameter count: ~LLM param count + HRM param count. E.g. LLaMA 13B has ~13e9. If HRM adds 50M, trivial percentage.

FLOPs (forward): A single Transformer block does ~$2TD^2$ flops (approx). So LLM forward is $\approx 2N_L BTD^2$. If we do **R** reasoning iterations through an HRM-like module of comparable size to say K layers, that adds $2RBTD^2$ (if using full Transformer attention each iter). HRM may have smaller effective T or D (e.g. operate on summary states), but worst-case it's like extra layers.

For example, LLM 13B (24 layers, D=4096, 16 heads) on T=1024: ~*224*B*1024*4096^2 flops per forward (about $2 * 24 * (4096^2) * 1024 \approx 2 * 24 * 16.7e6 * 1024 \approx 819e9 \approx 8.2e11$ FLOPs per sample!). Times B=1 yields ~0.82e12 FLOPs. This is huge, but practical GPUs do it in tens of ms.

If we add **R = 5** reasoning steps (say each HRM step is like 1 layer of size D), that adds ~$5 \times 2BTD^2 \approx 5 * (1/24) \times LLM_FLOPS \approx 0.2 \times LLM_FLOPS$. So +20%. If R=50, then ~10x cost (impractical). HRM paper reportedly used up to 8 cycles and 8 inner loops【8†L358-L362】, which is like R=~8, manageable.

Memory (Inference):

- **Activation memory:** We need to store key/value caches for attention (for generation) which is ~$B * T * D$ per layer. For 24 layers, ~24*B*T*D floats. With D=4096, T=1024, B=1: 24*1024*4096 ≈ 100M floats ≈ 400MB (FP32) or 200MB (FP16).
- **HRM memory:** The recurrent loops produce intermediate hidden states. If we do `R` steps with no grad (as HRM does for R-1 steps), we still need to hold `z_H` and `z_L`, but not backprop through them (detached). The final step's activations (for grad) add memory. Overall small relative to LLM.
- **Optimizer states:** Adam requires 2x parameters. If LLM+HRM have ~13B, Adam holds ~26B parameters (times 4 bytes = 104GB!). We actually use sharded or ZeRO to handle this in practice, otherwise impossible.
- **KV-cache:** In generation, we store all past keys/values for each layer. That's about (B*T*D) per layer, which becomes huge for long T. For T=2048, D=4096, per layer ~ (1*2048*4096=8.3M floats ~33MB FP16). For 24 layers ~792MB FP16.

Training memory: All activations for backward need to be stored. Even with gradient checkpointing, a large model may not fit non-distributed. The HRM's no-grad loops help reduce stored graph nodes, but still: each reasoning step's final layer activations for the backward pass. Memory is likely the primary bottleneck, necessitating techniques like activation checkpointing and large multi-GPU nodes.

Inference vs Training: Inference cost (FLOPs) scales linearly with R (reasoning steps) and T (generated length). Training cost scales with backprop (factor ~3x forward). HRM adds complexity because the backward pass can't easily share computations between steps.

Example Calculation: GPT-3 175B at seq=2048: ~350 TFLOPs per token generated (per OpenAI scaling law blog). If HRM adds 10% more compute, goes to ~385 TFLOPs. If HRM used R=10 steps, 100% more compute. For R=50, 500% more, likely infeasible.

Hence, an HRM-style model *must* keep R small or only use deep reasoning on sub-problems. For example, do not run full reasoning on every token in a conversation, only on specific query parts. Adaptive halting (Architecture D) can limit extra cost.

In summary, we expect:

- **1 reasoning step**: ~1× LLM cost.
- **5 steps**: ~1.2–1.5× cost (if HRM is small or reused).
- **10 steps**: ~2× cost.
- **50 steps**: ~5–6× cost (impractical for long dialogues).

Memory: increases modestly with R (storing a few extra hidden states) but main growth is storing KV cache and optimizer.

Optimization and Low-Level Performance

To make this hybrid model fast, we rely on optimized libraries and carefully choose what to implement manually:

- **Attention & MatMul:** Use **cuBLAS/cuDNN** or FlashAttention. Never write a naive attention loop in Python. We call existing kernels: e.g. `torch.nn.MultiheadAttention` with `flash_attn=True`. For multi-query attention, we use libraries like NVIDIA FasterTransformer. For MoE, use Megatron/Deepspeed's expert parallel kernels.

- **FlashAttention (DAO-AILab)**【25†L780-L788】: Integrate the official FlashAttention kernels (CUDA/Triton) so that Q*K and softmax are fused, drastically reducing memory traffic. This is crucial for long contexts. The HRM code requires FlashAttention (see its install instructions)【6†L245-L253】, so it's essential.

- **Triton Kernels:** For any custom heavy compute (like a fused Transformer block, or the HRM iteration itself if it can be expressed as a kernel), consider Triton. For example, one could implement an entire HRM inner loop block in Triton to fuse operations. Also useful for specialized activations (GELU, RMSNorm).

- **CUDA (custom):** Possibly write a CUDA kernel if Triton is insufficient. For instance, if designing a new type of reasoning module that does convolution on hidden states, implement in .cu.

- **Quantization:** Use existing libraries (TensorRT, ONNX Runtime). We should quantize mainly inference: map weights to INT8 and use INT8 matmul kernels on GPU/TPU. Some frameworks support GPT-int8 inference (NGC Triton).

- **Mixed Precision:** Use FP16 or BF16 in training (PyTorch AMP) to double throughput on Ampere/Hopper GPUs.

- **Data Pipeline:** Use fast tokenizers (Hugging Face Tokenizers in Rust/C++). For streaming datasets, use webdataset or TFRecord to avoid Python overhead.

- **Multi-GPU:**

- **Tensor Parallelism (GPipe/Megatron):** Split model weights across GPUs (e.g. split heads or hidden dim). Implement via frameworks (Megatron-LM, DeepSpeed).
- **Pipeline Parallelism:** If we have distinct modules (LLM vs HRM), pipeline them across GPUs (e.g. GPU1 does LLM layers, GPU2 does HRM layers) to overlap operations. Rare but possible.

- **Communication:** Use NCCL (in C++) for all-reduce of gradients. Python triggers this via `torch.distributed`.

- **Which to leave to frameworks:**

- Don't re-implement matrix multiply or conv in C++ by hand – rely on PyTorch's ATen.
- Only custom implement if a new op that PyTorch doesn't have efficiently (e.g. a special recurrence or gating mechanism).
- Use existing libraries: e.g. NVIDIA's cuDNN LSTM if we had RNN (though here we use Transformer).
- For multi-node training, PyTorch DDP handles the collective ops.

In short, **optimize only the new pieces**: likely the HRM recurrence and any new attention variants. Everything else (embedding, linear layers, etc.) use built-in C/CUDA. Use techniques like operator fusion (Torch 2.0 compiler) where possible.

Inference Architecture

At inference time (serving the model to users), we have to manage the **user input → reasoning → output** flow, often interactively:

1. **User Input:** Raw text query. Tokenize to IDs. (BPT token IDs).
2. **Encoding:** Pass tokens through shared Transformer encoder (or feed into decoder), obtaining hidden context `H`.
3. **Reasoning Decision:** Decide if we should do reasoning before answering. We might always do at least 1 reasoning loop, but the ACT controller can decide to stop after any step based on Q-values 【15†L1119-L1124】.
4. **HRM/Latent Loop:** Run the HRM loop on `H`. That may run a variable number of iterations, updating `z_H` and `z_L`. If halting early (ACT), we break. Each iteration uses the high-level and low-level modules【17†L1088-L1097】. All intermediate states are latent (not output to user).
5. **Language Generation:** Once reasoning stops, feed `z_H_final` into the decoder head. If this is an encoder-decoder, the decoder generates tokens while attending to `z_H_final`. If it's decoder-only, we insert `z_H_final` into the token stream (e.g. as the output of an initial prompt) and begin auto-regressive decoding. Typically, we'd append a sentinel token (like `<answer>`) and let the model generate.
6. **Token-by-Token Generation:** Generate tokens one at a time (or in parallel via sampling methods). Use KV-cache to speed up: after producing first token, store its key/values. When continuing, only new query (last token) needs computing, while old K/Vs are in cache. The HRM hidden state may be constant across all outputs, so we simply reuse it.

7. **Streaming Output:** The generation can be streamed to the user as tokens become available (with batching if needed).
8. **End Condition:** Stop when EOS token or max length. The reasoning controller doesn't intervene here (it already halted and passed final state).

Special considerations:

- The HRM's **internal state** (`z_H` , `z_L`) is **not directly exposed**. It only influences the language head. After answering, the next user query presumably resets the state (we do not maintain persistent reasoning memory across turns, unless we want a stateful assistant).
- **Batch Inference:** If multiple queries, we can batch them through the Transformer. The HRM loops for each in parallel, but ACT may halt each one at different steps. We'd manage a boolean `halted[i]` per example (like HRM's carry) and only continue loops for those still running 【15†L1169-L1177】.
- **Speculative Decoding:** Optionally use techniques like speculative decoding【34†L24-L33】: e.g., generate multiple tokens in draft from the LLM and refine them using an ensemble (though this is advanced and outside core architecture).
- **Quantized Inference:** Convert weights to INT8 and use quantized kernels for huge speed-ups (especially on NVidia TensorRT or on CPU with X86-VNNI). Many LLMs deploy INT8 now with minimal quality loss. HRM reasoning modules may be smaller, but we could quantize them too if stable.
- **Memory:** During inference, we hold the Transformer weights and HRM weights (plus caches). The HRM cache (`z_H` , `z_L`) is small. We still need space for the KV cache if generation length is large.

When reasoning occurs: We assumed reasoning happens *before generation*. One could also consider doing some reasoning *between tokens* (Architecture C variant). For example, the model generates one token, then does another HRM iteration, then next token. This interleaving could allow reasoning after seeing each word, but complicates implementation (storing and restoring hidden states). It's more complex than needed. It's simpler to do all reasoning first.

Thus our inference architecture: the model "thinks" internally via HRM (on the input), then "speaks" via Transformer decoding.

Distributed Training

For large-scale training (billions of parameters, massive data), we employ multi-GPU/multi-node strategies:

- **Data Parallelism (DDP):** Each GPU holds a full model copy (or sharded if using ZeRO). Each gets a mini-batch, computes gradients, and then all GPUs `all-reduce` to sync weights. We do this with PyTorch DDP (which uses NCCL under the hood). Works well up to ~hundreds of GPUs.

- **Tensor Parallelism:** For extremely large models (100B+), split individual layers' weights across GPUs. E.g. split each linear's weight matrix by columns or heads (like Megatron-LM). Then matmuls become collective ops. If using experts (MoE), could parallelize experts. This is more complex and requires framework support.

- **Pipeline Parallelism:** If the model can be partitioned into stages (e.g. first 12 layers on GPU0, last 12 on GPU1), one can pipeline micro-batches through. In practice, often combined with tensor parallelism (e.g. DeepSpeed's 3D parallelism).

- **Fully Sharded Data Parallel (FSDP):** If GPU memory is tight, shard parameters/optimizer across GPUs (ZeRO). This is useful when training large transformer models. We'd likely use ZeRO Stage 2/3 so that each GPU only stores a fraction of model states.

- **Expert Parallelism:** If using MoE, experts can be sharded across GPUs (expert parallel). Also need to all-reduce within expert outputs.

In our code, we would integrate with PyTorch Lightning or Fairseq / HuggingFace training loops that support DDP and ZeRO. The HRM part is small, so the main scale is defined by LLM size. We must ensure **random seeds, batching, and data sharding** are done consistently (PyTorch DDP sets each worker's seed from a base).

Hardware: Modern clusters have NVLink or InfiniBand. Use NVLink within node for faster all-reduce, and NCCL over IB across nodes.

Monitoring: Use tools (W&B, TensorBoard) to track gradient norms, learning rates, etc. Save checkpoints periodically (e.g. every N updates) and keep best on dev set.

Failure Modes and Risks

- **Reasoning Loops Diverge:** If the iterative HRM loop fails to converge to a stable solution (e.g., it oscillates or falls into trivial cycles), the model may output garbage. *Detection:* Monitor hidden-state changes; if they increase or cycle, halt. *Mitigation:* Limit max steps (like HRM's `halt_max_steps` 【15†L1192-L1197】). Use gating (ACT) to force stop.

- **Overthinking / Inconsistency:** The model might "think" too much and produce inconsistent outputs on similar inputs. *Mitigation:* Regularize halting (penalty per step). Provide examples of varying difficulty.

- **Training Instability:** The recursive architecture may cause exploding/vanishing gradients. HRM avoids some by detaching intermediate steps, but still risk. *Mitigation:* Use gradient clipping, careful initialization (e.g. trunc-normal as HRM does【23†L994-L1002】), and small learning rates on HRM part.

- **Representation Collapse:** The model might ignore one of the modules (e.g. treat HRM as identity). Check that the reasoning module indeed changes hidden state (ablation studies). Use auxiliary losses to encourage usage if needed.

- **Hallucinations:** Typical LLM problem. Fusion may reduce this if reasoning grounds answers, but could also hallucinate reasoning steps. *Mitigation:* Include fact-check data in training, or retrieval augmentation.

- **Inference Cost:** If HRM decides many steps for hard queries, latency could blow up. Must cap steps and maybe degrade to faster mode if too slow. Possibly run HRM asynchronously (user sends query, system computes reason offline, then responds).

- **Dataset Contamination:** If training data for HRM includes answers to test puzzles (like ARC), model might memorize. Use proper train/test splits, data dedup, and evaluate on held-out problems.

- **Compatibility:** Some Transformer tricks (like multi-query attention) may conflict with the HRM's need to examine all heads. We must verify that any speed hack doesn't degrade reasoning.

- **Memory Limits:** Large models + loops may exceed GPU memory. Use memory-efficient attention, checkpointing.

- **Gradient Flow:** The HRM uses no-grad on intermediate steps【17†L1097-L1105】; if not handled properly, gradients might be blocked incorrectly. Ensure correct use of `torch.no_grad()`.

- **Complexity/Bug:** Combining two complex architectures increases risk of integration bugs (shape mismatches, indexing errors, gating logic flaws). Use extensive tests (unit tests in `tests/`) and sanity checks on a small model.

Detecting failures involves monitoring training curves, ablation (turn off reasoning to see difference), and testing on known benchmarks (both language and reasoning tasks). Mitigations often involve ablation (removing/adapting components), normalization layers (to stabilize), and fallbacks (e.g. if reasoning fails, just answer with LLM plain output).

Minimal Prototype Design

A minimal educational prototype would include:

- A simple Transformer (few layers, small D) implemented in PyTorch.
- A toy HRM: maybe one high-level LSTM and one low-level LSTM (rather than full transformer blocks) in a loop for a fixed small number of steps.
- A trivial tokenizer (split on spaces, small vocab).
- A tiny dataset (few example puzzles).
- Python code orchestrating forward/backward.

This prototype demonstrates: how to unroll an inner loop, use `torch.no_grad()`, compute a halt decision, and output tokens. Code could be <200 lines. It would run on CPU or 1 GPU. No need for multi-GPU or optimized kernels. It just shows concept.

Scalable Research Implementation

As we scale up, we add:

- Replace simple LSTMs with full Transformer blocks and FlashAttention.

- Increase model size (more layers, bigger D).
- Use real tokenizer (BPE) and large datasets (bookcorpus, etc.).
- Integrate with distributed training (DDP, ZeRO).
- Write custom CUDA kernels if needed (e.g. for fused multi-step updates).
- Add logging, config management (e.g. Hydra), robust exception handling.
- Optimize data pipeline (sharding, caching).
- Expand test suite with integration tests (train a few steps, generate known answer).
- Infrastructure (Docker images, CI/CD for tests).
- Profiling tools to find bottlenecks.

At full scale, each module (Transformer, HRM loop) will be built from battle-tested libraries (PyTorch, NVIDIA kernels) and only glue logic in Python.

Example Code

Below are illustrative snippets (not full modules) to convey the key ideas:

```
# Example: Simplified Transformer Block (PyTorch)
class TransformerBlock(nn.Module):
    def __init__(self, D, num_heads, ff_dim):
        super().__init__()
        self.attn = torch.nn.MultiheadAttention(embed_dim=D,
num_heads=num_heads, batch_first=True)
        self.ff = nn.Sequential(
            nn.Linear(D, ff_dim), nn.GELU(), nn.Linear(ff_dim, D)
        )
        self.norm1 = nn.LayerNorm(D)
        self.norm2 = nn.LayerNorm(D)
    def forward(self, x, attn_mask=None):
        # Self-attention (causal mask assumed set externally)
        attn_out, _ = self.attn(x, x, x, attn_mask=attn_mask)
        x = x + attn_out
        x = self.norm1(x)
        # Feed-forward
        ff_out = self.ff(x)
        x = x + ff_out
        x = self.norm2(x)
        return x
```

```
# Example: HRM Inner Loop (high-level pseudocode)
class HRMLayer(nn.Module):
    def __init__(self, D):
        super().__init__()
        self.high = torch.nn.GRUCell(D, D)  # high-level state update
        self.low  = torch.nn.GRUCell(D, D)  # low-level state update
```

```
        self.q_head = nn.Linear(D, 2)       # predict halt/continue logits

    def forward(self, z_h, z_l, combined_input):
        # One step of HRM iteration
        z_l_new = self.low(combined_input, z_l)
        z_h_new = self.high(z_l_new, z_h)
        # Compute halting Q-values from high-level state
        q_logits = self.q_head(z_h)
        return z_h_new, z_l_new, q_logits
```

```
# Example: HRM wrapper with ACT (very simplified)
class HierarchicalReasoner(nn.Module):
    def __init__(self, D, max_steps=5):
        super().__init__()
        self.max_steps = max_steps
        self.hrm_layer = HRMLayer(D)

    def forward(self, z_h, z_l, input_emb):
        # z_h, z_l initial (maybe zeros or pooled from encoder)
        halted = False
        for step in range(self.max_steps):
            combined = z_h + input_emb  # inject input at each step
            z_h, z_l, q_logits = self.hrm_layer(z_h, z_l, combined)
            # Deciding halt based on Q
            if q_logits[0] > q_logits[1]:  # naive: logit0=halt, logit1=cont
                halted = True
            if halted:
                break
        return z_h  # output final state

# Integrating with Transformer for one token generation
class HybridModel(nn.Module):
    def __init__(self, vocab_size, D):
        super().__init__()
        self.embed = nn.Embedding(vocab_size, D)
        self.transformer = nn.ModuleList([TransformerBlock(D, num_heads=8,
ff_dim=4*D) for _ in range(6)])
        self.hrm = HierarchicalReasoner(D, max_steps=3)
        self.head = nn.Linear(D, vocab_size)  # language head

    def forward(self, input_ids):
        x = self.embed(input_ids)  # (B,T,D)
        # pass through transformer backbone
        for layer in self.transformer:
            x = layer(x)
        # Pool to get initial state, e.g. first token
```

```
        z_h = x[:,0,:]
        z_l = z_h.clone()
        # Run HRM reasoning
        z_final = self.hrm(z_h, z_l, x[:,0,:])
        # Produce logits for next token (for simplicity, use first position)
        logits = self.head(z_final)
        return logits
```

This code sketches how Python orchestrates modules and uses PyTorch operations. The **actual numeric ops** (embedding lookup, matmuls in MultiheadAttention, linear layers) run in C++/CUDA inside PyTorch. If we needed a custom attention, we would insert calls to the FlashAttention op (e.g. `flash_attn_forward`) in place of `torch.nn.MultiheadAttention`.

For a **CUDA extension example**, one might write (in `kernels/my_attention.cu`):

```
// C++ pseudocode: custom kernel wrapper
#include <torch/extension.h>
torch::Tensor my_fast_attention(torch::Tensor Q, torch::Tensor K, torch::Tensor
V) {
    // Launch CUDA kernel that computes attention using blocks and shared
memory...
    // (Actual code would use cudaLaunchKernel and so on.)
    return output;
}
PYBIND11_MODULE(TORCH_EXTENSION_NAME, m) {
    m.def("my_fast_attention", &my_fast_attention, "Fast Attention");
}
```

And in Python:

```
import torch
import my_attention  # compiled C++/CUDA module
class MyAttn(nn.Module):
    def forward(self, Q, K, V):
        return my_attention.my_fast_attention(Q, K, V)
```

For **Triton**, one would write a kernel in a `.py` file using the Triton API:

```
import triton
import triton.language as tl

@triton.jit
def fused_lin_elu(OUT, IN, weight, bias, M, N, BLOCK: tl.constexpr):
    # A simple example: linear + ELU fused kernel
```

```
    row = tl.program_id(0)
    cols = tl.arange(0, BLOCK)
    idx = row * N + cols
    x = tl.load(IN + idx)
    w = tl.load(weight + cols * M + row)
    y = x * w
    if bias is not None:
        y += tl.load(bias + cols)
    # ELU activation
    out = tl.where(y>0, y, tl.exp(y)-1)
    tl.store(OUT + idx, out)
```

Then call this in Python with the Triton launcher.

These code examples illustrate the **structure and interfaces**: Python modules call lower-level kernels, and training loops just do `loss.backward()` with no need to manage those details.

Open Problems

Despite the design above, many questions remain unsolved:

- **Emergent vs explicit reasoning:** Can a purely architectural fusion like this truly surpass chain-of-thought prompting? It's unclear if explicit recurrence is necessary for all tasks. Future work should test if HRM+LLM actually improves reasoning on e.g. math or code.
- **Efficient iteration:** Best ways to allow iterative thinking without huge cost is open. ACT-style halting helps, but training it stably is tricky. Methods to learn how many steps without supervision are not fully solved.
- **Modular vs monolithic:** How independent can reasoning and language be? E.g. can we share attention weights between modules? The tension between specialization and parameter sharing is an area for exploration.
- **Model interpretability:** A hierarchical model might be more interpretable (explicit steps) but validating intermediate reasoning remains hard. Designing losses or probes to ensure the model reasons correctly is unsolved.
- **Memory and retrieval:** The approach here doesn't explicitly handle vast knowledge (RAG needed). Integrating dynamic retrieval in HRM style (deciding what to look up mid-reasoning) is an open research area.
- **Scaling laws for reasoning:** Most scaling laws focus on loss vs compute. It's unknown how much additional capacity yields better explicit reasoning. Does bigger always help, or do we hit algorithmic limits?
- **Human-like thinking strategies:** HRM is inspired by brain hierarchies, but the brain likely uses attention, memory, and other modules. Incorporating more neuroscience or cognitive patterns (episodic memory, attention switching) is speculative.

In summary, building an HRM+LLM fusion system is an ambitious engineering challenge. The path outlined here is a hypothesis driven by current research, but only thorough experimentation will reveal which design

choices truly work. We have pointed to the primary research sources and sketched a roadmap; implementing, testing, and iterating on this design is an open frontier.

Research References

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- Sanh et al., *"DistilBERT"*, 2019 (Model distillation).
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All citations above are to peer-reviewed papers, official implementations, or documented engineering reports. The design proposals here are based on synthesizing these sources with engineering reasoning; speculative components are noted as such.
